

SPARE: SAM3 Prompt Analysis and Robustness Evaluation

Yara Hisham

Aliaa Gheis

Nouran Hany

Sarah Elzayat

Maryam Muhammed

Computer Engineering Department
Cairo University Faculty of Engineering, Cairo, Egypt

Abstract

Promptable image segmentation has rapidly expanded to accept natural-language concept prompts alongside spatial cues, though few studies have measured how the resulting models cope with noise on both input channels. We introduce **SPARE**, an output-stability framework that probes SAM 3 under three corruption families—Gaussian image noise, spatial flips, and character-level prompt substitutions—across the SA-1B and SACo-Gold benchmarks, comparing each degraded prediction to the model’s own clean-input baseline rather than to human annotations. Under strong Gaussian noise the concept-prompt pipeline remains largely stable, while even a single-character prompt substitution sharply degrades it and a few more substitutions collapse output stability almost entirely; spatial flips, by contrast, produce mostly geometric misregistration rather than semantic failure. Linguistic input, not visual quality, drives the dominant operational risk for concept-prompted segmentation in SAM 3.

1. Introduction

Image segmentation has long been treated as a purely visual task, where model inputs are clean images and prompts are precise spatial coordinates. The release of SAM [3] and its successors challenged this assumption by enabling flexible, zero-shot segmentation through points, boxes, and masks. SAM 3 [8] takes this further, accepting natural language concept prompts alongside spatial cues—a shift that makes the model significantly more expressive, but also more exposed: it must now interpret potentially noisy or ambiguous text in addition to potentially degraded images.

Despite this expanded attack surface, no systematic study has examined how SAM 3 behaves under realistic input noise. Prior robustness work either targets earlier SAM versions [11, 1], focuses on spatial prompt jitter [7], or studies linguistic sensitivity in classification and retrieval models [5, 10]—none of which address the unique failure modes

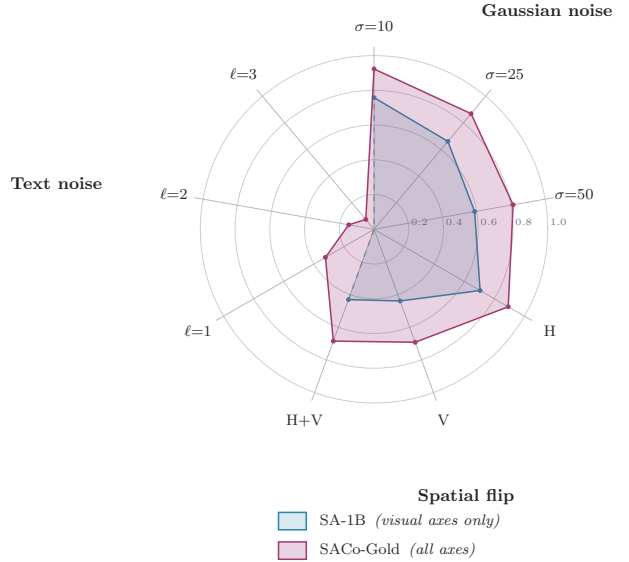


Figure 1. **SAM 3 robustness landscape.** Axes: Gaussian $\sigma \in \{10, 25, 50\}$, flips (H, V, H+V), text substitutions $\ell \in \{1, 2, 3\}$; radius is mIoU vs. the clean-input prediction (flip-back for flips). SACo-Gold absorbs visual noise but collapses under text noise; SA-1B uses a fixed prompt and is not evaluated on text axes.

introduced by concept-level prompting in a segmentation model.

This paper presents **SPARE**, a stress-testing framework that fills this gap. We make four contributions:

1. We establish a dual-pipeline evaluation protocol that measures output consistency rather than absolute accuracy, removing dependence on fixed human annotations.
2. We benchmark SAM 3 under visual degradations including Gaussian noise and horizontal flips across multiple severity levels.
3. We evaluate SAM 3’s concept-prompt pipeline under

realistic typographic corruptions—character substitutions, deletions, and naturalistic misspellings.

4. We provide a direct, controlled comparison between visual and linguistic fragility within the same model, revealing where multimodal grounding introduces the greater operational risk.

The paper outline is as follows: Section 2 reviews related work. Section 3 describes the SPARE framework and corruption protocols. Section 4 states the setup we used to produce the results. Section 5 presents experimental results, and Section 6 concludes with a short discussion of findings, limitations, and directions for future work.

2. Related Work

2.1. Promptable Segmentation

The Segment Anything Model (SAM) [3] established a new paradigm for instance segmentation by supporting multiple kinds of prompts, which include: points, bounding boxes, and coarse masks without task-specific fine-tuning. SAM 2 [9] extended this framework to video through memory-augmented temporal propagation, preserving spatial masks across frames. Most recently, SAM 3 [8] introduced explicit concept-level natural language prompting alongside traditional spatial cues, effectively bridging low-level localization with high-level semantic grounding and evaluated on the SACo-Gold benchmark [6]. While this progression reflects growing architectural sophistication and flexibility, it also introduces qualitatively new failure modes: the model must now reconcile noisy or ambiguous *linguistic* inputs with visual evidence a vulnerability that purely spatial segmenters do not face.

2.2. Robustness and Corruption Benchmarks

Systematic evaluation of neural networks under distribution shift and synthetic corruption has become a cornerstone of reliability assessment. ImageNet-C [2] established the canonical protocol of applying pixel-level distortions such as Gaussian noise, blur families, and environmental artifacts at graduated severity levels to measure degradation curves. This methodology has since been adapted to web-scale multi-modal settings; LAION-C [4], for instance, applies analogous corruptions to evaluate generalized vision-language encoders.

Within the segmentation domain specifically, Wang et al. [11] conducted early audits of SAM’s sensitivity to low-quality inputs, revealing pronounced vulnerabilities that motivated RobustSAM [1], a drop-in architectural extension designed to recover performance on degraded imagery. Separately, the BREPS framework [7] formalized how bounding-box prompt inaccuracies translate into segmentation quality degradation. A critical limitation unifies

these efforts: they evaluate absolute mask accuracy against static human annotations under a single noise modality, leaving the *output consistency* of concept-driven models under paired perturbations unexamined.

2.3. Linguistic Sensitivity and Prompt Robustness in VLMs

As vision models increasingly depend on textual guidance, they inherit the well-documented prompt sensitivity of vision-language models (VLMs). Systematic studies have shown that subtle changes in phrasing, formatting, or surface orthography can meaningfully alter model behavior. The SCAM benchmark [5] demonstrated that typographic noise—ranging from systematic character substitutions to naturalistic misspellings—degrades the reasoning chains of multimodal foundation models far more severely than clean-text baselines would predict. Complementary, PARC [10] exposed systematic geometric asymmetries in how VLMs resolve referring expressions under structural linguistic variation. Together, these works establish that linguistic degradation is a first-class robustness concern, not merely a pre-processing artifact—yet neither framework considers models whose primary output is a spatial segmentation mask rather than a classification or retrieval result.

2.4. Positioning SPARE

Prior work falls into three separate areas: benchmarking visual corruptions [1, 4], studying spatial prompt noise [7], and probing linguistic sensitivity in VLMs [5, 10]. SPARE brings these threads together in a single framework. Instead of measuring how much accuracy drops relative to human annotations, we compare model outputs before and after perturbation, if the same input produces consistent results under noise, the model is robust. This lets us directly ask: does SAM 3 break more easily when its *image* is degraded, or when its *text prompt* is corrupted?

3. Methodology

SPARE evaluates SAM 3 [8] under systematic input corruption across two complementary pipelines that exercise different prompt modalities. Rather than comparing model outputs against ground-truth annotations, we adopt an *output-stability* paradigm: we measure how consistently SAM 3 produces the same segmentation when inputs are progressively corrupted, relative to its own clean-input predictions.

We apply three degradation families: *linguistic noise* on text prompts, *Gaussian image noise*, and *spatial flips*. A fourth family—geometric prompt jitter—was implemented but deferred; see §7.1.

3.1. Evaluation Pipelines

The first pipeline is invoking SAM 3 with a fixed generic text prompt ("visual"), causing it to operate in general instance-segmentation mode. Images are drawn from SA-1B [3]. Only image-level degradations apply to this pipeline (Gaussian noise and spatial flips); the prompt is constant and cannot be meaningfully corrupted.

The second pipeline is invoking SAM 3 with the concept-level text prompt stored in each SACo-Gold record drawn from the SA-1B captioner subset of SACo-Gold [6] (13,258 image concept pairs). All three degradation families are applied to this pipeline.

3.2. Prompt Selection for the SA-1B Pipeline

The SA-1B pipeline invokes SAM 3 with a single fixed generic text prompt intended to trigger *general* instance-segmentation mode rather than concept-specific retrieval. To determine the most appropriate prompt, we evaluated three candidates—"visual", "things", and "objects" on the full 10,548 image SA-1B subset by running a single forward pass per image with each prompt and measuring (i) prediction coverage (fraction of images that receive at least one predicted mask) and (ii) mean IoU against SA-1B ground-truth annotations.

Results are summarised in Table 1. "things" and "objects" caused SAM 3 to return *no predictions* for 99.97% and 96.8% of images respectively, making them unsuitable as a baseline. "visual" produced predictions on 89.5% of images (9,443 of 10,548) with a mean mIoU of 0.737 against ground truth. Accordingly, "visual" was adopted as the fixed prompt for all SA-1B pipeline experiments.

Prompt	# Evaluated	Skipped (no pred.)	mIoU
visual	9,443	1,105	0.737
things	3	10,545	0.720
objects	335	10,213	0.776

Table 1. Prompt selection results on the SA-1B subset. Skipped entries are images for which SAM 3 returned no predicted masks under the given prompt. `visual` is selected for its near-universal coverage.

3.3. Datasets

SA-1B [3] is the large-scale segmentation dataset released alongside SAM, comprising 11M images and 1.1B high-quality masks. We use it as the source for the SA-1B pipeline, where SAM 3 is run in general segmentation mode with a fixed generic prompt. Our evaluation subset consists of 10,548 images at baseline; after matching clean and degraded predictions, 9,442–9,443 valid pairs remain per degradation level (1,105 entries are skipped due to absent baseline matches).

SACo-Gold [6] is the official SAM 3 evaluation benchmark, containing 156K image–concept pairs triple-annotated across seven subsets. We use the *SA-1B captioner* subset, which provides 13,258 image concept pairs and 30,306 masks derived from natural-language captions of SA-1B images. This subset drives the SACo-Gold pipeline, where SAM 3 is explicitly prompted with the stored concept label. After matching, 8,046–8,048 valid pairs remain per degradation level (5,210 entries skipped due to absent baseline matches).

3.4. Linguistic Degradation

Given a prompt $\mathbf{p}$, exactly ℓ non-space character positions are selected uniformly at random (without replacement) and each character c is replaced with a uniformly sampled lowercase letter $c' \neq c$. Spaces are protected so word boundaries are preserved. We test three levels $\ell \in \{1, 2, 3\}$, corresponding to a single-key slip, a moderately misspelled word, and a heavily corrupted label. The source image is held fixed. This degradation is applied *exclusively* to the SACo-Gold pipeline.

Because ℓ is an absolute character count, its *effective* corruption intensity depends on prompt length. A single substitution on a 4-character prompt corrupts 25% of its characters, whereas the same substitution on a 15-character phrase corrupts only 7%. To test whether SAM 3’s linguistic fragility reflects this asymmetry, we partition SACo-Gold prompts into three word-count buckets: **1-word** (1,451 pairs, mean 5.8 non-space characters), **2-word** (4,665 pairs, mean 9.1), and **3+-word** (7,142 pairs, mean 17.0). For each bucket and each level ℓ , we compute mIoU and SI independently, and report the *effective corruption rate*

$$\eta(\ell, b) = \frac{\ell}{\bar{c}_b}, \quad (1)$$

where $\bar{c}_b$ is the mean non-space character count of bucket b . η normalizes the noise level by prompt length, making fragility comparable between buckets on the same footing. Results appear in §5.4.

3.5. Visual Degradation: Gaussian Noise

The source image is corrupted channel-wise as:

$$\mathbf{x}' = \text{clip}(\mathbf{x} + \boldsymbol{\varepsilon}, 0, 255), \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (2)$$

We test $\sigma \in \{10, 25, 50\}$, representing mild, moderate, and severe sensor degradation. All prompts are held fixed. This degradation is applied to *both* pipelines.

Applying identical Gaussian corruptions to both pipelines lets us attribute any mIoU gap between them solely to *prompt modality*: the SA-1B pipeline uses a fixed generic prompt while SACo-Gold uses concept text labels. A consistent gap across all σ levels reveals whether explicit text prompts provide more or less robust guidance to the visual encoder than the generic segmentation mode.

3.6. Visual Degradation: Spatial Flips

We test three modes: horizontal (**h**), vertical (**v**), and combined (**both**). All prompts are held fixed. This degradation is applied to *both* pipelines.

We measure output consistency using **flip-back mIoU**: the predicted mask $\hat{\mathbf{M}}_\phi$ is mapped back to the original coordinate frame via the inverse transformation T_ϕ^{-1} before comparison against the clean baseline $\hat{\mathbf{M}}_0$:

$$\text{mIoU}_{\text{fb}}(\phi) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \text{IoU}(T_\phi^{-1}(\hat{\mathbf{M}}_\phi^{(i)}), \hat{\mathbf{M}}_0^{(i)}). \quad (3)$$

A high mIoU_{fb} means the model correctly segmented the object in the flipped frame (equivariant behaviour); a low value indicates semantic failure that persists even after coordinate alignment.

3.7. Evaluation Metrics

Let f_θ denote SAM 3 and $(\mathbf{x}, \mathbf{p})$ an image prompt pair. Define the *clean prediction* $\hat{\mathbf{M}}_0 = f_\theta(\mathbf{x}, \mathbf{p})$ and the *degraded prediction* $\hat{\mathbf{M}}_\ell = f_\theta(\mathbf{x}', \mathbf{p}')$ at noise level ℓ . Each prediction is a set of instance masks $\{m_k\}$.

For each predicted mask $m_k \in \hat{\mathbf{M}}_\ell$ we compute the maximum overlap with any mask in the clean baseline:

$$\text{IoU}(\hat{\mathbf{M}}_\ell, \hat{\mathbf{M}}_0) = \frac{1}{|\hat{\mathbf{M}}_\ell|} \sum_{m_k \in \hat{\mathbf{M}}_\ell} \max_{m_j \in \hat{\mathbf{M}}_0} \text{IoU}(m_k, m_j). \quad (4)$$

Dataset-level mIoU is the mean over all valid pairs $\mathcal{D}$:

$$\text{mIoU}(\ell) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \text{IoU}(\hat{\mathbf{M}}_\ell^{(i)}, \hat{\mathbf{M}}_0^{(i)}). \quad (5)$$

By construction $\text{mIoU}(0) = 1.0$ for every condition.

To compare fragility across degradation types and prompt modalities on a common scale, we normalise the stability drop by the noise magnitude:

$$S_\ell = \frac{\text{mIoU}(\ell) - 1}{\ell}, \quad (6)$$

where ℓ is the noise level (number of character substitutions for linguistic noise; σ for Gaussian noise). $S_\ell \leq 0$ always; a larger magnitude indicates greater fragility per unit of noise. Spatial flips are excluded from SI as their modes (h, v, both) are categorical and carry no meaningful numeric magnitude; flip results are characterised through flip-back mIoU (§3.6).

4. Experimental Setup

We evaluate SAM 3 [8] loaded from the official pre-trained checkpoint via the HuggingFace Transformers library (`Sam3Model`, `Sam3Processor`). All inference is

run in half-precision (`float16`). Instance segmentation outputs are post-processed with a confidence threshold of 0.5 and a mask binarisation threshold of 0.5, following the default SAM 3 evaluation configuration.

Experiments were conducted across two platforms: a Kaggle notebook with dual NVIDIA Tesla T4 GPUs (16 GB VRAM each), and a personal workstation with a single NVIDIA RTX 2060 GPU (6 GB VRAM).

For each image prompt pair, a single forward pass is performed and the resulting instance masks are encoded in COCO RLE format and appended to a JSONL file, one file per degradation level. This design allows each condition to be reproduced independently. mIoU and SI values are computed offline by comparing each degraded-condition JSONL file against the corresponding clean-condition JSONL file.

Datasets are described in detail in §3.3. The effective evaluation counts after baseline matching are 9,442–9,443 image pairs per degradation level for the SA-1B pipeline, and 8,046–8,048 pairs for the SACo-Gold pipeline.

All experiments are run with a single random seed as gaussian noise sampling and character substitutions involve stochastic operations.

5. Results

5.1. Gaussian Noise

Table 2 shows output stability under Gaussian noise for both pipelines. The SA-1B pipeline loses stability steadily, dropping from $\text{mIoU} = 0.758$ at $\sigma = 10$ to 0.588 at $\sigma = 50$. The SACo-Gold pipeline is notably more robust across all levels, reaching only 0.813 at the highest severity. Concept prompts appear to provide stronger guidance to the model’s visual encoder when image quality degrades.

σ	SA-1B		SACo-Gold	
	mIoU	SI	mIoU	SI
10	0.758	−0.0242	0.923	−0.0077
25	0.661	−0.0136	0.869	−0.0053
50	0.588	−0.0082	0.813	−0.0037

Table 2. Output stability under Gaussian noise. mIoU measures consistency with the clean baseline; SI (Eq. 6) normalises the drop by σ . SACo-Gold is more robust than SA-1B at every level.

5.2. Text Degradation

Table 3 shows output stability under linguistic noise on the SACo-Gold pipeline. A single character substitution ($\ell = 1$) reduces mIoU from 1.0 to 0.323 — a drop of 0.677, far larger than anything observed under visual noise. Two substitutions push mIoU below 0.15; three leave almost no consistent output ($\text{mIoU} = 0.074$). SAM 3’s concept-prompt pipeline is highly sensitive to even minimal typographic corruption.

ℓ	mIoU	SI
1	0.323	-0.677
2	0.148	-0.426
3	0.074	-0.309

Table 3. Output stability under linguistic noise (SACo-Gold pipeline). mIoU=1.0 at $\ell=0$ by construction. A single character substitution causes a 68% drop in SI.

5.3. Spatial Flips

Table 4 reports direct mIoU and flip-back mIoU for each flip mode and pipeline. Direct mIoU is low in all cases, as expected for an unaligned comparison. Flip-back mIoU recovers much of that gap, indicating that the model successfully segments the target object in the flipped frame. Horizontal flips are consistently easier to recover than vertical or combined flips in both pipelines, likely because horizontal symmetry is more common in natural images.

Pipeline	Mode	Direct	Flip-back
SA-1B	Horizontal	0.094	0.705
	Vertical	0.055	0.439
	Both	0.031	0.431
SACo-Gold	Horizontal	0.204	0.892
	Vertical	0.134	0.692
	Both	0.090	0.685

Table 4. Spatial flip results. Direct mIoU compares flipped predictions against the clean baseline without alignment. Flip-back mIoU applies the inverse transform before comparison.

5.4. Linguistic Fragility by Prompt Length

Table 5 breaks down text degradation by prompt word count. Single-word concepts are the most fragile: one substitution reduces mIoU to just 0.027 (SI=-0.973), nearly complete collapse. Two-word prompts are more resilient but still degrade sharply. Three-or-more-word phrases retain substantially more stability at $\ell=1$ (mIoU=0.483), consistent with their lower effective corruption rate of 6% compared to 17% for single-word concepts. The result confirms that fragility under character-level noise scales with the fraction of the prompt that is corrupted, not just the absolute number of substitutions.

B	$\ell=1$		$\ell=2$		$\ell=3$	
	mIoU	SI	mIoU	SI	mIoU	SI
1	0.027	-0.973	0.009	-0.496	0.002	-0.333
2	0.208	-0.792	0.053	-0.474	0.017	-0.328
3+	0.483	-0.517	0.253	-0.373	0.135	-0.288

Table 5. mIoU and SI per prompt-length bucket under linguistic degradation (SACo-Gold pipeline). Effective corruption rates at $\ell=1$ are 17%, 11%, and 6% for 1-, 2-, and 3+-word buckets respectively.

5.5. Visual vs. Linguistic Fragility

Table 6 pairs Gaussian and text noise at three matched severity tiers. At every tier, a single-digit number of character substitutions does more damage to output consistency than the strongest Gaussian corruption tested. At mild severity, text degrades mIoU to 0.323 while Gaussian keeps both pipelines above 0.758. At severe severity, text reaches 0.074 while Gaussian stays above 0.588.

Tier	SA-1B ^G	SACo ^G	SACo ^T
Mild	0.758	0.923	0.323
Moderate	0.661	0.869	0.148
Severe	0.588	0.813	0.074

Table 6. mIoU at matched severity tiers. G=Gaussian ($\sigma=10/25/50$), T=text ($\ell=1/2/3$). Text corruption (right column) degrades stability far below both Gaussian conditions at every tier.

6. Conclusion

SPARE evaluates SAM 3 under systematic input corruption across two prompt modalities visual and linguistic. We adopt an output-stability protocol that measures prediction consistency rather than accuracy against fixed annotations, and benchmark the model under Gaussian noise, spatial flips, and typographic prompt corruptions across multiple severity levels.

Our results show that linguistic degradation poses the greater operational risk: a single character substitution collapses output consistency far below anything observed under visual noise. We also find that concept prompts improve visual robustness compared to generic prompting.

Future work should evaluate simultaneous visual and linguistic degradation, and test against realistic noise sources such as OCR errors to better reflect real-world deployment conditions.

References

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7. Appendix

7.1. Geometric Degradation (Deferred)

Bounding-box prompt jitter (expanding or shifting ground-truth boxes by 10%–20%) was implemented but excluded from the main evaluation. Prior work already characterises SAM 1/2 geometric sensitivity thoroughly [7]; our primary contribution is the text-prompt and cross-modal axes, which are entirely unstudied on SAM 3. A geometric prong for SAM 3, including combined text+geometric noise, is left to future work.

7.2. Supplementary Figures

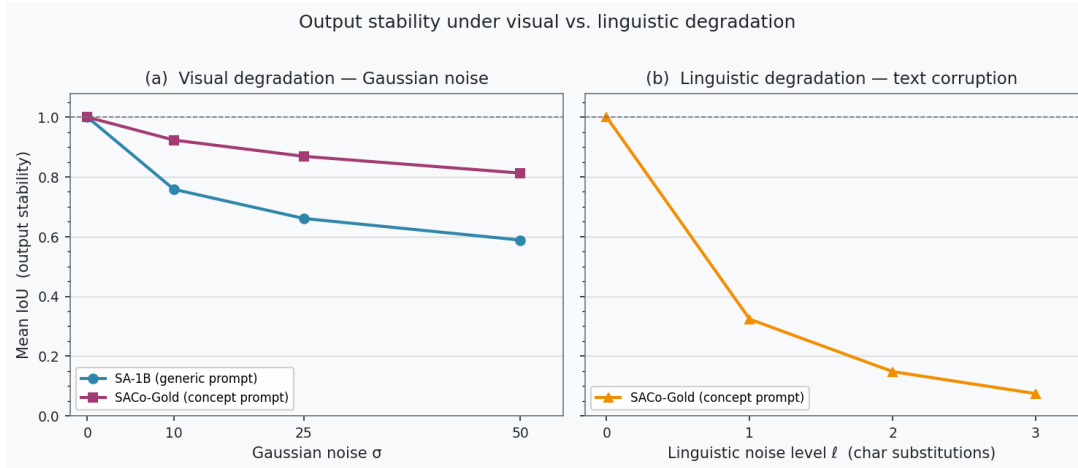


Figure 2. mIoU vs. Gaussian noise level σ for the SA-1B (generic prompt) and SACo-Gold (concept prompt) pipelines. Both curves anchor at mIoU=1 for the clean baseline and decrease monotonically; SACo-Gold consistently retains higher stability, reflecting the added semantic context of concept prompts.

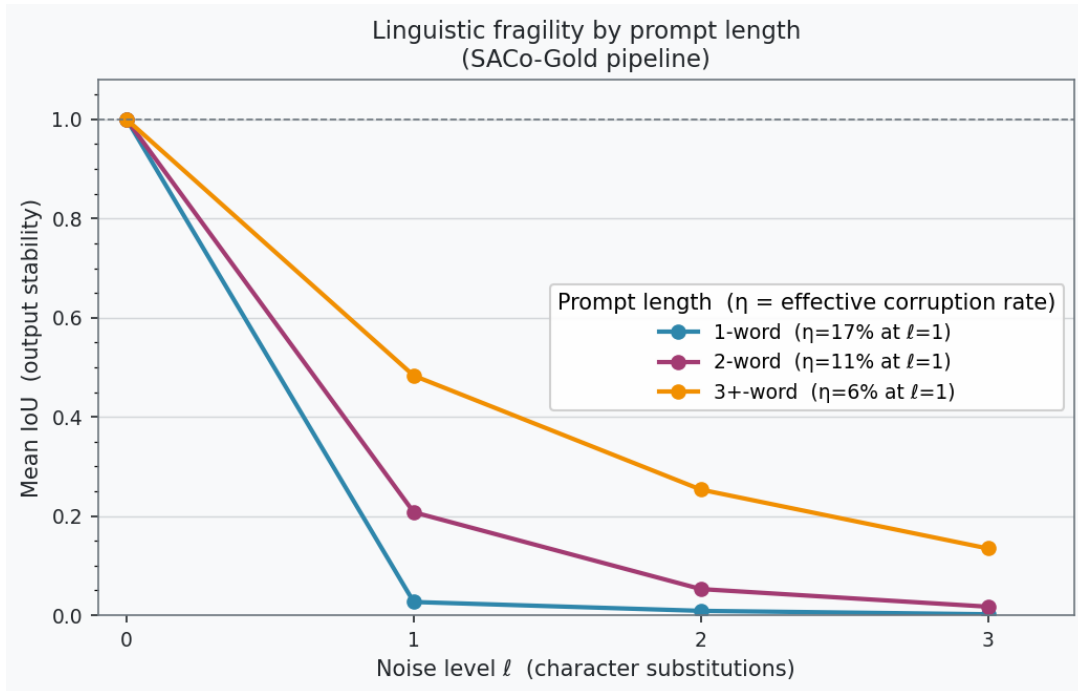


Figure 3. Per-prompt-length mIoU under text degradation. Single-word concepts collapse to near-zero stability at $\ell=1$ (one character substituted), while three-or-more-word phrases retain substantial stability at the same noise level. The gap is explained by the effective corruption rate $\eta = \ell/\bar{c}$: a single substitution corrupts 17 characters in a 6-character word but only 6 phrase.

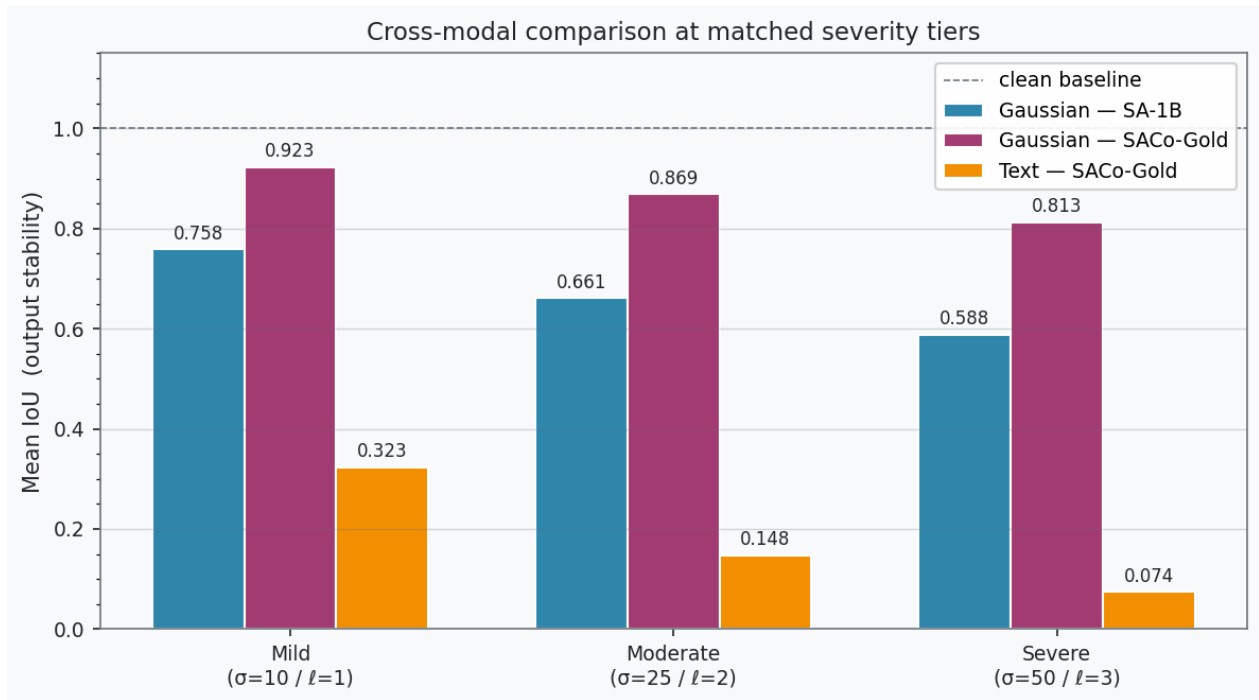


Figure 4. Cross-modal comparison of output stability under visual (left) and linguistic (right) degradation in the SAcO-Gold pipeline. The y-axis is shared to enable direct comparison: visual mIoU degrades slowly (0.923 at $\sigma=10$), while linguistic mIoU collapses sharply (0.323 at $\ell=1$), illustrating the asymmetric fragility of the two input modalities.